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Parallel Block-Iterative Reconstruction Algorithms for Binary Tomography

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Abstract

A convex programming approach to binary tomographic image reconstruction in noisy environments is proposed. Conventional constraints are mixed with new constraints on the sinogram. A convex objective is then minimized over the resulting feasibility set via a parallel block-iterative method. The new constraints involve noise-based confidence regions and a binarity-promoting total variation constraint.

Keywords: Binary tomography, block-iterative algorithm, convex constraints, convex programming, image reconstruction, total variation.

1 Introduction

The tomographic reconstruction problem is to estimate a multidimensional signal $\overline{x}$ in a Hilbert space $\mathcal{H}$ from lower dimensional measurements of its line integrals. In computerized tomography, the supports of the signals are discretized over a bounded domain and $\mathcal{H}$ is the usual Euclidean space $\mathbb{R}^N$

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(using standard lexicographic ordering, an N-voxel signal is represented as an N-point vector). A popular approach for solving this problem is to pose it as a convex feasibility problem of the form

$$\text{Find } \overline{x} \in \bigcap_{i=1}^{m} S_i, \tag{1}$$

where $(S_i)_{1\leq i\leq m}$ are closed convex sets in $\mathbb{R}^N$ arising from prior knowledge (bounds, support information, spectral information, information about the noise corrupting the measurements, etc.) and the discrete line integral (line sum) measurements [3,4,14,20]. This set theoretic approach to tomographic image reconstruction goes back to [12]; further developments can be found in [16,17,19,22]. In some instances it may be justified on physical grounds to seek a feasible image which is optimal in some sense. The problem then assumes the form

$$\text{Find } \overline{x} \in S = \bigcap_{i=1}^{m} S_i \text{ such that } \varphi(\overline{x}) = \inf \varphi(S), \tag{2}$$

where $\varphi\colon \mathbb{R}^N \to \left]-\infty, +\infty\right]$ is a convex function. The advantage of such convex programming formulations is to allow for the modeling of a wide range of prior information in the reconstruction process and, at the same time, to benefit from the availability of powerful solution algorithms; see [1,3,5,6,7] and the references therein.

In discrete tomography, the range of the signal to be reconstructed is known to be a finite set of values, for instance the set $\{0, 1\}$ in binary tomography. This additional information is of paramount importance and it has profound consequences on the theoretical and practical aspects of the reconstruction problem [15]. As underlined in [15], classical computer tomography algorithms do not perform well in the presence of few (say 10 or less) views (we do not employ the standard term "projection," as it will be reserved to describe the best metric approximation from a convex set). Consequently, they are not directly applicable in discrete tomography, where such low numbers of views are common. Furthermore, classical algorithms do not exploit nor enforce the discrete nature of the original signal.

In this paper, we shall focus on the reconstruction of binary images. In recent years, several approaches have been proposed to solve this problem; see [11,13,15,21] and the references therein. We propose a formulation of the binary tomographic reconstruction problem as a convex program of the form (2) and to solve it via the parallel block-iterative method of [7]. Some attractive features of this algorithm are:

- its ability to use of approximate (subgradient) projections onto the constraint sets, which makes it possible to incorporate a wide range of nonlinear constraints in the form of convex inequalities;
- its ability to perform parallel processing of variable blocks of constraints;
- its potential for accelerated convergence through the use of extrapolated over-relaxations.

Since the set of binary images is nonconvex, our first task will be to find pertinent *convex* constraints that will promote the binary nature of the image. Thus, the image produced by the algorithm will be relatively close to binary, which will minimize the number of errors incurred by the final binarization step.

The remainder of the paper is as follows. In Section 2, we review the parallel block-iterative algorithm that will be employed to solve the convex program (2). In Section 3, we discuss the construction of constraints for binary tomography. The new constraints are confidence regions that are based on statistical attributes of the noise perturbing the sinogram, and a total variation constraint aiming at promoting the binary nature of the reconstructed image. In Section 4, we describe several applications of this convex programming framework to binary image reconstruction in the presence of noisy measurements, limited observation angle range, and limited number of views. A few remarks conclude the paper in Section 5.

2 Proposed algorithm

Throughout this paper, the signal space containing the original signal $\overline{x}$ is the standard Euclidean space $\mathbb{R}^N$, with scalar product $\langle \cdot \mid \cdot \rangle$ and norm $\|\cdot\|$.

To solve (2), we shall use the parallel block iterative algorithm described in [7], where the framework of [6] was adapted to problems with quadratic objective functions. Although more general strictly convex objectives φ can be used [6], we restrict our attention to quadratic functions in this paper, as they lead to particularly simple implementations.

Let

$$\varphi\colon \mathbb{R}^N \to \mathbb{R}\colon x \mapsto \langle R(x-r) \mid x-r \rangle, \tag{3}$$

where $r \in \mathbb{R}^N$ and $R \in \mathbb{R}^{N\times N}$ is a positive definite symmetric matrix. In addition, we suppose that the constraint sets $(S_i)_{1\leq i\leq m}$ in (2) assume the form

$$(\forall i \in \{1,\ldots,m\}) \;\; S_i = \{x \in \mathbb{R}^N \mid f_i(x) \leq \delta_i\}, \tag{4}$$

where $(f_i)_{1\leq i\leq m}$ are convex functions from $\mathbb{R}^N$ to $\mathbb{R}$ and $(\delta_i)_{1\leq i\leq m}\in\mathbb{R}^m$ are such that $S=\bigcap_{i=1}^m S_i\neq\varnothing$. Recall that, under these assumptions, for every $x\in\mathbb{R}^N$, f_i admits at least one subgradient at x, i.e., a point $g_i\in\mathbb{R}^N$ such that

$$(\forall y\in\mathbb{R}^N)\quad \langle y-x\mid g_i\rangle+f_i(x)\leq f_i(y). \tag{5}$$

The set of all subgradients of f_i at x is the subdifferential of f_i at x and is denoted by $\partial f_i(x)$; if f_i is differentiable at x, then $\partial f_i(x)=\{\nabla f_i(x)\}$. Moreover, the subgradient projection G_ix of x onto S_i is obtained by selecting an arbitrary $g_i\in\partial f_i(x)$ and setting

$$G_ix=\begin{cases}x+\dfrac{\delta_i-f_i(x)}{\|g_i\|^2}g_i, & \text{if } f_i(x)>\delta_i;\\ x, & \text{if } f_i(x)\leq\delta_i.\end{cases} \tag{6}$$

Note that computing G_ix requires only the availability of a subgradient (a gradient in the differentiable case) of f_i at x and is therefore much more economical than computing the exact projection P_ix, as the latter amounts to solving a constrained quadratic minimization problem.

The algorithm proposed in [7] to solve (2)–(3) constructs a sequence $(x_n)_{n\in\mathbb{N}}$ of approximate solutions as follows.

Algorithm 1

1. *Fix* $\varepsilon\in\left]0,1/m\right[$. *Set* $x_0=r$ *and* $n=0$.
2. *Take a nonempty index set* $I_n\subset\{1,\ldots,m\}$.
3. *Set* $z_n=x_n-\lambda_nR^{-1}u_n$, *where:*
   - *for every* $i\in I_n$, $p_{i,n}=G_ix_n$;
   - *the weights* $(\omega_{i,n})_{i\in I_n}$ *lie in* $[\varepsilon,1]$ *and* $\sum_{i\in I_n}\omega_{i,n}=1$;
   - $u_n=x_n-\sum_{i\in I_n}\omega_{i,n}p_{i,n}$;
   - $\lambda_n\in[\varepsilon L_n,L_n]$, *where*

$$L_n=\begin{cases}\dfrac{\displaystyle\sum_{i\in I_n}\omega_{i,n}\|p_{i,n}-x_n\|^2}{\langle R^{-1}u_n\mid u_n\rangle}, & \text{if } \max\limits_{i\in I_n}\left(f_i(x_n)-\delta_i\right)>0;\\ 1/\|R^{-1}\|, & \text{otherwise.}\end{cases}$$

4. *Set* $\pi_n=\langle R(x_0-x_n)\mid x_n-z_n\rangle$, $\mu_n=\langle R(x_0-x_n)\mid x_0-x_n\rangle$, $\nu_n=\langle R(x_n-z_n)\mid x_n-z_n\rangle$, *and* $\rho_n=\mu_n\nu_n-\pi_n^2$.

5. *Set*

$$x_{n+1} = \begin{cases} z_n, & \text{if } \rho_n = 0,\ \pi_n \geq 0; \\ x_0 + \left(1 + \dfrac{\pi_n}{\nu_n}\right)(z_n - x_n), & \text{if } \rho_n > 0,\ \pi_n \nu_n \geq \rho_n; \\ x_n + \dfrac{\nu_n}{\rho_n}(\pi_n(x_0 - x_n) + \mu_n(z_n - x_n)), & \text{if } \rho_n > 0,\ \pi_n \nu_n < \rho_n. \end{cases}$$

6. *Set* $n = n + 1$ *and go to Step 2.*

The following convergence result is an application of [7, Theorem 16].

Theorem 2.1 *Suppose that there exists a strictly positive integer* M *such that*

$$(\forall n \in \mathbb{N}) \quad \bigcup_{k=n}^{n+M-1} I_k = \{1, \ldots, m\}. \tag{7}$$

Then every sequence $(x_n)_{n\in\mathbb{N}}$ *generated by Algorithm 1 converges to the unique solution to (2)–(3).*

Some comments about this result and Algorithm 1 are in order.

- Condition (7) is satisfied in particular if $I_n \equiv \{1, \ldots, m\}$, i.e., all the sets are activated at each iteration. In general, (7) allows for variable blocks of sets to be used, which offers great flexibility in terms of parallel implementation (see [5] for examples). More details on the importance of block-processing for task scheduling on parallel architectures will be found in [3]. Further flexibility is offered by the fact that the relaxations and the weights can vary at each iteration.
- Algorithm 1 activates the constraints via subgradient projections rather than exact projections. The former are significantly easier to implement than the latter, as they require only the computation of subgradients (gradients in the differentiable case). Analytically complex constraints can therefore be incorporated in the recovery algorithm and processed at low cost.
- The parameter L_n is always at least equal to $1/\|R^{-1}\|$ [7, Proposition 12] and can attain large values. Choosing λ_n large (e.g., equal to L_n) usually yields faster convergence.

In [7], it was shown that, in order to reduce the computational load of the iterations, Algorithm 1 can be implemented as follows. This implementation requires only one application of the matrices R and R^{-1}.

Algorithm 2

1. *Fix* $\varepsilon \in]0, 1/m[$. *Set* $x_0 = r$ *and* $n = 0$.
2. *Take a nonempty index set* $I_n \subset \{1, \ldots, m\}$.
3. *For every* $i \in I_n$, *set* $a_i = -f_i(x_n)g_i/\|g_i\|^2$, *where* $g_i \in \partial f_i(x_n)$, *if* $f_i(x_n) > \delta_i$; $a_i = 0$ *otherwise.*
4. *Choose weights* $(\omega_i)_{i\in I_n}$ *in* $[\varepsilon, 1]$ *adding up to 1. Set* $v = \sum_{i\in I_n} \omega_i a_i$ *and* $L = \sum_{i\in I_n} \omega_i \|a_i\|^2$.
5. *If* $L = 0$, *set* $x_{n+1} = x_n$ *and exit iteration. Otherwise, set* $b = x_0 - x_n$, $c = Rb$, $d = R^{-1}v$, *and* $L = L/\langle d \mid v \rangle$.
6. *Choose* $\lambda \in [\varepsilon L, L]$ *and set* $d = \lambda d$.
7. *Set* $\pi = -\langle c \mid d \rangle$, $\mu = \langle b \mid c \rangle$, $\nu = \lambda \langle d \mid v \rangle$, *and* $\rho = \mu\nu - \pi^2$.
8. *Set* $x_{n+1} = \begin{cases} x_n + d, & \text{if } \rho = 0 \text{ and } \pi \geq 0; \\ x_0 + (1 + \pi/\nu)d, & \text{if } \rho > 0 \text{ and } \pi\nu \geq \rho; \\ x_n + \dfrac{\nu}{\rho}(\pi b + \mu d), & \text{if } \rho > 0 \text{ and } \pi\nu < \rho. \end{cases}$
9. *Set* $n = n + 1$ *and go to Step 2.*

We note that, at Step 3, if the projection $P_i x_n$ of x_n onto S_i is easy to compute, one can set $f_i = d_{S_i}$ (the distance function to S_i), and (6) yields $a_i = P_i x_n - x_n$.

3 Construction of constraint sets

3.1 Standard constraints

The first standard constraint arises from the fact that pixel values are nonnegative and have known maximal value. After normalization, the corresponding set is

$$S_1 = [0, 1]^N. \tag{8}$$

In the context of binary tomography, S_1 is simply the convex hull of the set $\{0, 1\}^N$. A second common piece of a priori information in tomography is the knowledge of the support K of the body being studied. The associated set is

$$S_2 = \{x \in \mathbb{R}^N \mid x \cdot 1_K = x\}, \tag{9}$$

where 1_K is the characteristic function of K. It is clear that both S_1 and S_2 are closed convex subsets of $\mathbb{R}^N$. Further details on such standard constraint sets can be found in [4], including the closed-form expression of the projection

onto these sets.

3.2 Constraint based on noise properties

The sinogram is the image under the Radon transform of the original signal $\overline{x}$. The portion of the sinogram corresponding to a given observation angle θ will be referred to as a *view*. The observed data consists of q noisy views $(s_i)_{1\leq i\leq q}$ at angles $(\theta_i)_{1\leq i\leq q}$. For every $i \in \{1,\dots,q\}$, we let L_i be the restriction of the Radon transform for a fixed angle θ_i. In other words, the ith measurement is

$$s_i = L_i\overline{x} + w_i, \tag{10}$$

where w_i is the noise component corrupting the observation. Now, let M be the length of the signal s_i and let $(e_k)_{1\leq k\leq M}$ be the canonical basis of $\mathbb{R}^M$. If no noise were present, (10) would confine $\overline{x}$ to the intersection of the Mq hyperplanes $\{x \in \mathbb{R}^N \mid \langle L_i x \mid e_k\rangle = \langle s_i \mid e_k\rangle\}$. These sets were used in the early ART reconstruction technique [12]. In the presence of noise, the hyperplanes must be replaced by hyperslabs of the form $\{x \in \mathbb{R}^N \mid |\langle L_i x - s_i \mid e_k\rangle| \leq \delta_{i,k}\}$ [14], where $\delta_{i,k}$ is a bound on the random variable $|\langle w_i \mid e_k\rangle|$. Due to the complexity of the noise process, obtaining accurate bounds is a delicate task. Moreover, in most models, the noise has unbounded support and $\delta_{i,k}$ must be determined according to some confidence level. Since the number of hyperslabs is large, the confidence level on each set must be very close to one in order to yield a reasonable global confidence level [8]. As a result, the hyperslabs will be wide and, hence, not very restrictive.

Using an approach developed in [10], a wide range of statistical constraints can be formed for each of the q views. Indeed, (10) gives $L_i\overline{x} - s_i = w_i$. Hence, the residual signal $L_i x - s_i$ associated with an estimate x of $\overline{x}$ should be constrained to be statistically consistent with every known attribute (e.g., moment, periodogram, etc) of the noise. For instance, an estimate σ_i^2 of the energy $\|w_i\|^2$ of the noise may be available from physical considerations or past experience with tomographic reconstructions of similar objects. Under mild assumptions on the samples of the noise vector w_i, one can form the set

$$S_{i+2} = \{x \in \mathbb{R}^N \mid \|L_i x - s_i\|^2 \leq \delta_i\}, \tag{11}$$

where δ_i depends on σ_i and some statistical confidence level [10]. As discussed in [5], the projection of a signal x onto this set requires an iterative procedure, while the subgradient projection is easily derived from (6) to be

$$G_i x = \begin{cases} x + \dfrac{\delta_i - \|L_i x - s_i\|^2}{2\|L_i^\top (L_i x - s_i)\|^2} L_i^\top (L_i x - s_i), & \text{if } \|L_i x - s_i\|^2 > \delta_i; \\ x, & \text{if } \|L_i x - s_i\|^2 \leq \delta_i. \end{cases} \tag{12}$$

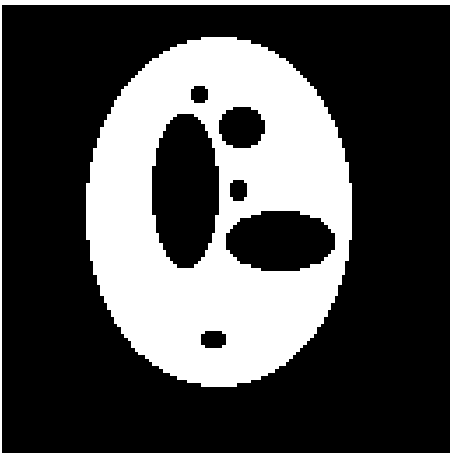


Fig. 1. Original image.

3.3 Binarity-promoting constraint

As discussed in the Introduction, constrained image recovery algorithms, such as those of [5,7], have many attractive features. However, they are limited to problems with convex constraints. As a result, since the set of binary images is nonconvex, the binarity constraint cannot be enforced directly in such frameworks. Furthermore, the trivial convex relaxation (8) will not properly enforce binarity and we must find a more effective means to promote binarity through a convex constraint.

In recent years, total variation has emerged as an effective tool to recover piecewise smooth images in variational image recovery. This approach was initiated in [18] in the context of denoising problems and has since been used in various restoration problems. Recently, it has also been applied in certain variational continuous computerized tomography problems [2,23]. In all these approaches, total variation appears in the objective of a minimization problem. Such nondifferentiable problems are not easy to solve and they offer limited potential in terms of incorporating constraints.

In [9], it was observed that, in many problems, the total variation $\mathrm{tv}(\overline{x})$ of the original signal, which measures the amount of oscillations, does not exceed some known bound ζ. This constraint, which is associated with the set

$$S_m = \{x \in \mathbb{R}^N \mid \mathrm{tv}(x) \leq \zeta\}, \tag{13}$$

appears to be particularly relevant in binary tomography, as it suppresses the oscillating components in the image, thereby forcing the creation of flat areas and promoting binarity. A important issue is of course the availability of

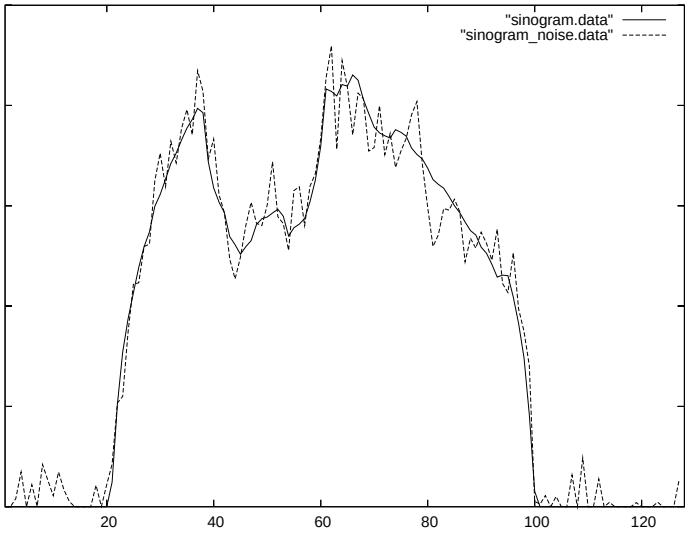


Fig. 2. Example of a sinogram view.

the parameter ζ in (13) [9]. In this respect, binary tomography places us on favorable grounds. Indeed, since the total variation of a binary image is simply the length of its contours (i.e., the sum of the perimeters of the elementary shapes), it can be estimated with good accuracy in certain typical problems from prior experiments or by sampling databases.

The total variation of a discrete image $x = [x^{i,j}] \in \mathbb{R}^{M\times M}$ is the convex function

$$\begin{aligned}\mathrm{tv}(x) = &\sum_{i=0}^{M-2}\sum_{j=0}^{M-2}\sqrt{|x^{i+1,j}-x^{i,j}|^2+|x^{i,j+1}-x^{i,j}|^2}\\ &+\sum_{i=0}^{M-2}|x^{i+1,M-1}-x^{i,M-1}|+\sum_{j=0}^{M-2}|x^{M-1,j+1}-x^{M-1,j}|.\end{aligned} \tag{14}$$

The subgradient projection $G_m x$ of a point x onto S_m can be found in [9].

4 Numerical simulations

The original 128×128 binary image $\overline{x}$ is shown in Fig. 1. This phantom displays features at different scales. As in [15], few projections are used in the experiments. Thus, while a typical computerized tomography reconstruction requires on the order of one hundred views, we shall use less than 10 views.

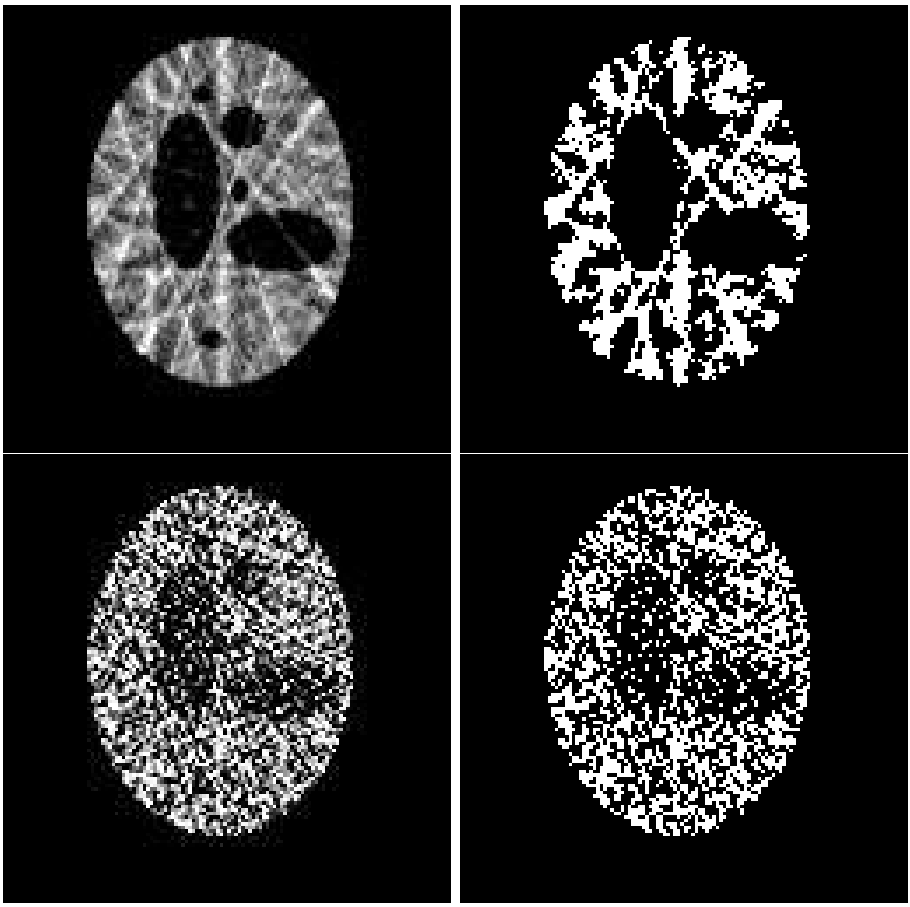


Fig. 3. Filtered back-projections with range and support constraints before (left) and after (right) binarization. Top: reconstruction with 10 noiseless views. Bottom: reconstruction with 10 noisy views (SNR = 0 dB).

4.1 Noise simulation

Noise modeling in computerized tomography is a research topic in its own right, and it is beyond the scope of the present paper to attempt to provide a precise model of the complex underlying physical phenomena. In our experiments, the noise corrupting the line-sum measurements is simulated by adding to the original image a Gaussian white noise w and by clipping to zero the possible negative values of the resulting sinogram. Indeed, a major source of noise is due to radiations being diffracted to the wrong detector. This can be seen as a positive perturbation on the wrong detector and a negative pertur-

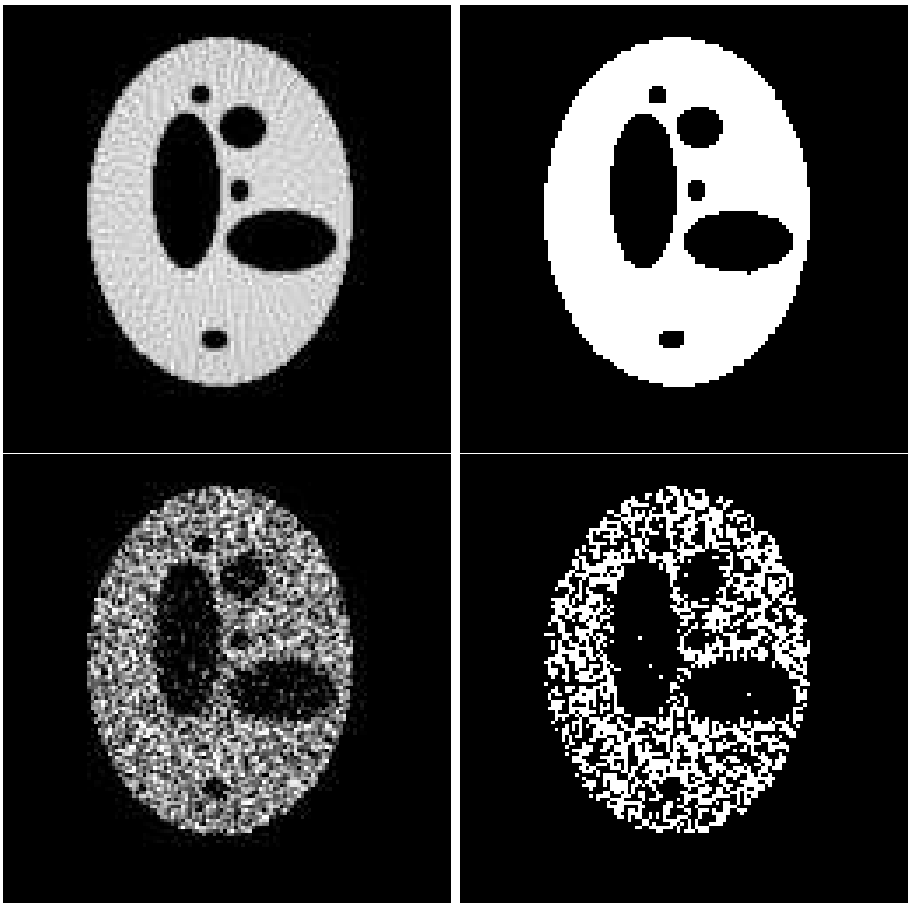


Fig. 4. Filtered back-projections with range and support constraints before (left) and after (right) binarization. Top: reconstruction with 50 noiseless views. Bottom: reconstruction with 50 noisy views (SNR= 0 dB).

bation on the originally intended detector. Since the resulting signal cannot be negative, the negative values are removed.

The signal-to-noise ratios (SNR) mentioned hereafter refer to the original image, i.e., to the expression $10\log_{10}(\|\overline{x}\|^2/\|w\|^2)$, expressed in decibels (dB). For instance, a SNR of 0 dB means that the energy of the simulated noise component w is the same as that of the original image $\overline{x}$. A typical noisy sinogram view corresponding to a 0 dB SNR is shown in Fig. 2.

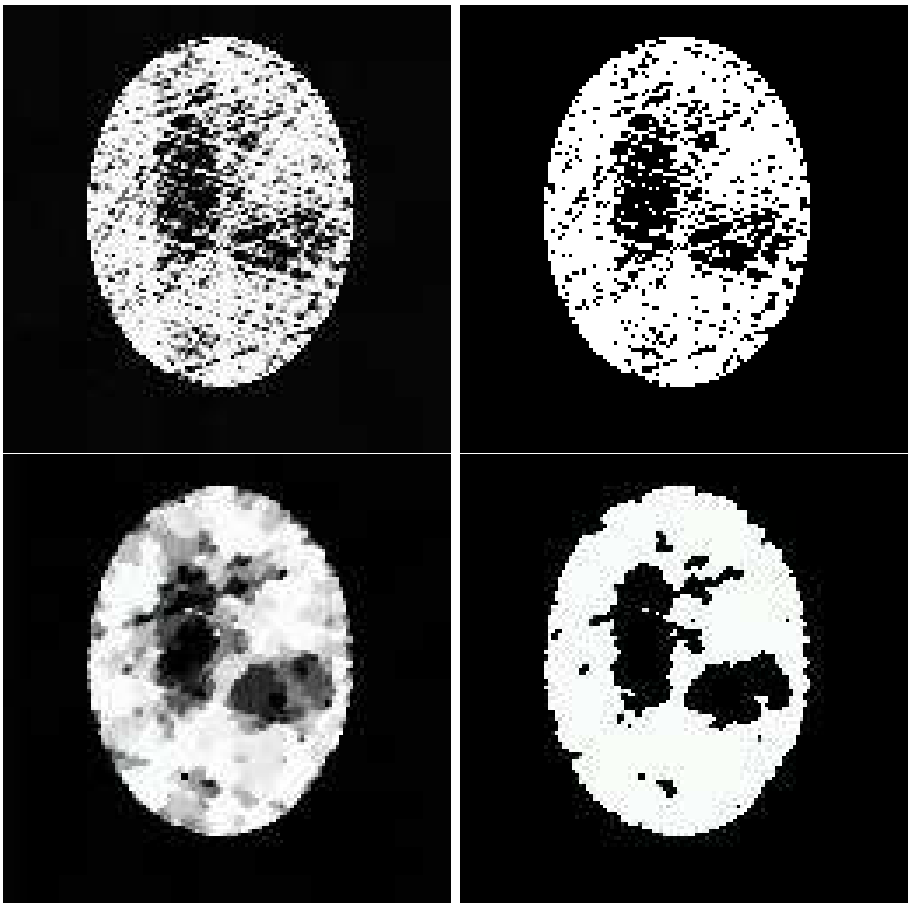


Fig. 5. Full range, 5 views, SNR = 0 dB, before (left) and after (right) binarization. Top: without the total variation constraint. Bottom: with the total variation constraint.

4.2 Binarization

The binarization process consists of mapping an image in $[0,1]^N$ into an image in $\{0,1\}^N$. The scheme we adopt here is straightforward: each pixel value is rounded 0 if the original value is less than 0.5, and to 1 otherwise. It is clear that binarization could be performed in a more sophisticated fashion, especially in the light of additional a priori information such as the energy of the original image.

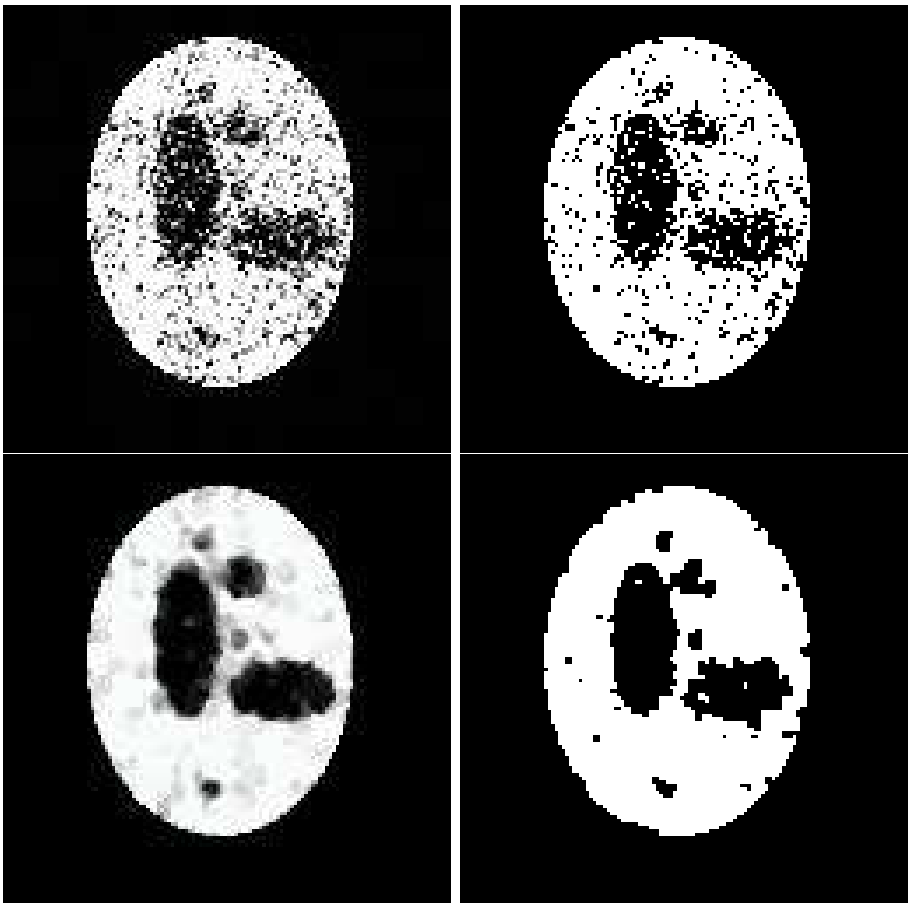


Fig. 6. Full range, 10 views, SNR = 0 dB, before (left) and after (right) binarization. Top: without the total variation constraint. Bottom: with the total variation constraint.

4.3 Filtered back-projection

Figs. 3 and 4 show the reconstruction obtained by projecting the standard filtered back-projection onto the set $S_1 \cap S_2$ (see (8) and (9)), i.e., by enforcing the range and support constraints.

4.4 Experimental setup

Algorithm 1 is implemented with $\varphi\colon x \mapsto \|x-r\|^2$ in (3) (hence R is the identity matrix), where r is the constrained filtered back-projected image described in

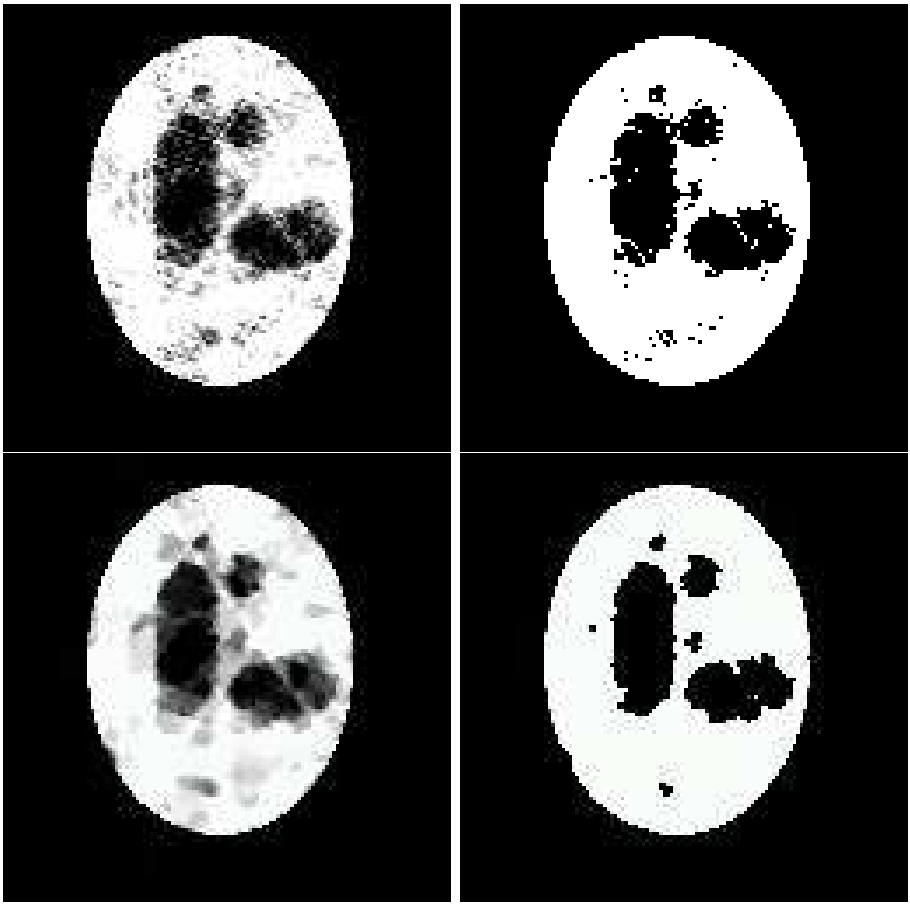


Fig. 7. Full range, 5 views, SNR = 10 dB, before (left) and after (right) binarization. Top: without the total variation constraint. Bottom: with the total variation constraint.

Section 4.3. In other words, we seek the best approximation to r from the feasibility set $S = \bigcap_{i=1}^{m} S_i$. The $m = q + 2$ constraint sets to be used are:

- S_1: pixel range, see (8).
- S_2: image support, see (9).
- $(S_{i+2})_{1\leq i\leq q}$: residual energy of the views, see (11).
- S_m: total variation, see (13).

To illustrate the benefit of using the total variation set S_m, we perform experiments with the sets $(S_i)_{1\leq i\leq m-1}$ and $(S_i)_{1\leq i\leq m}$.

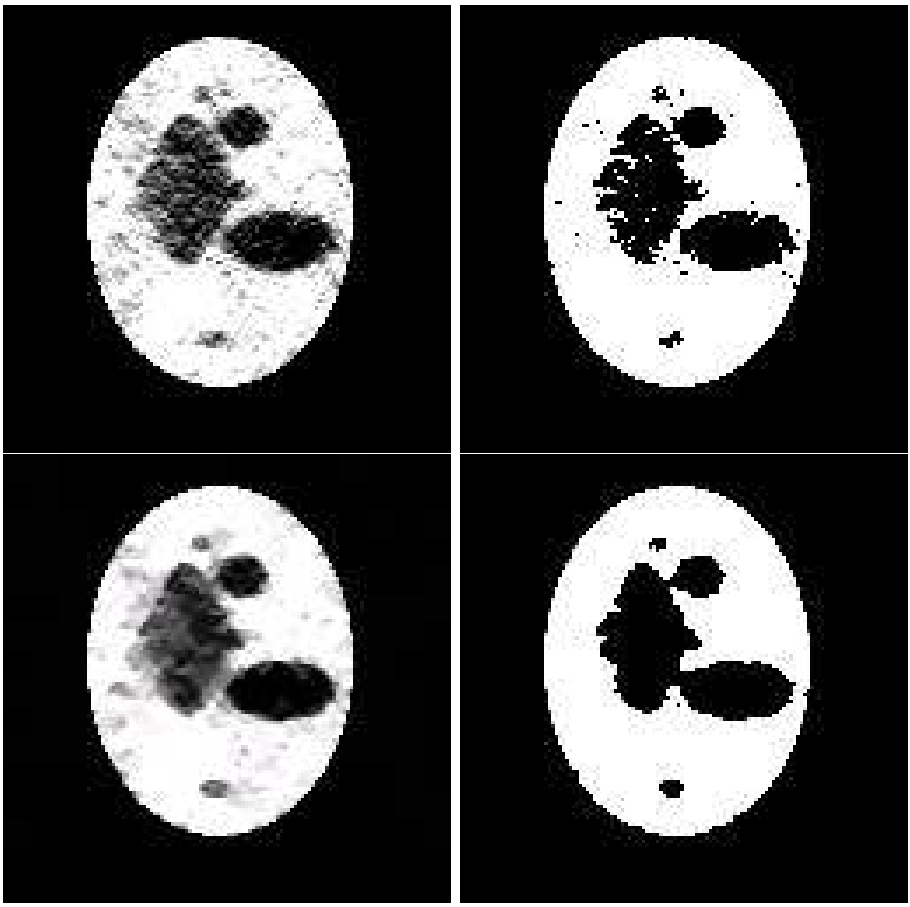


Fig. 8. 120 degrees limited range, 10 views, SNR = 10 dB before (left) and after (right) binarization. Top: without the total variation constraint. Bottom: with the total variation constraint.

Since they are easily computable in closed form, exact projections onto the sets S_1 and S_2 are performed. On the other hand, subgradient projections are used for the sets $(S_i)_{3\leq i\leq m}$ (see Sections 3.2 and 3.3).

4.5 Numerical results

We first consider *full range* problems, i.e., problems in which line integrals can be chosen along paths at angle ranging from 0 to 180 degrees. By contrast, in *limited angle* problems, the angles of the views will be confined to some range. In both scenarios, the views are uniformly spaced in the angle range.

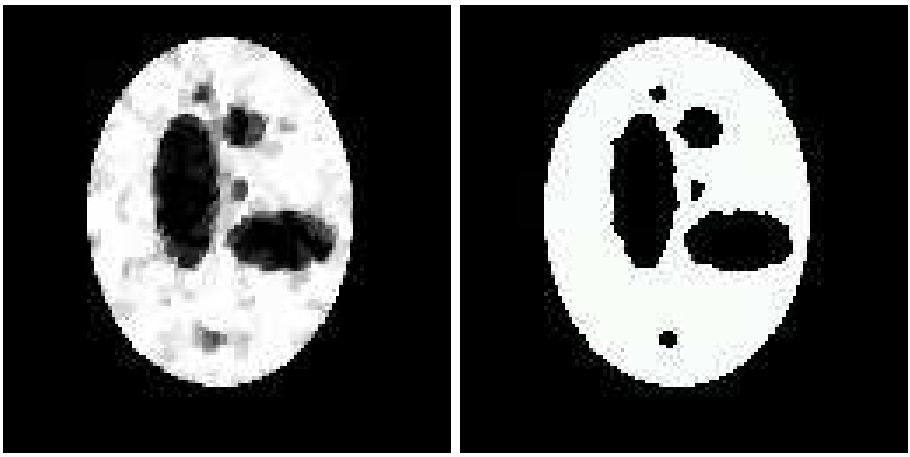


Fig. 9. Full range, 10 views, SNR = 10 dB, with the total variation constraint before (left) and after (right) binarization.

The experiments feature very few views and severe noise corruption.

Figs. 5–7 show the effectiveness of the total variation constraint in various situations. In Fig. 7, only 5 views are used in a 10 dB noise. Yet, all the features of the original image are reconstructed and few artifacts are created. When the total variation constraint is employed, even the small pattern at the bottom of the image is recovered.

Fig. 8 features a noisy limited angle problem. Here also, the improvement due to the total variation constraint is clear. Finally, Fig. 9 is an example of an almost perfect binary reconstruction from noisy data and only 10 views.

5 Concluding remarks

We have proposed a convex programming approach to the binary tomographic image reconstruction problem. Promising results have been obtained with a limited number of constraints and a quadratic objective function. The proposed algorithm can also handle a wide range of additional constraints as well more general objective functions. The exploration of such extensions is left for future work; in particular, additional constraints could be derived from a better understanding of the noise process. Likewise, while total variation seems to give good results, soft enforcement of binarity should be possible through alternative convex functionals.

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